

TOOL_BOX

Version 0.29 Functionality Guide

Menu Reference, Settings, Workspaces, Reporting, Utilities, and Administration

Program	Tool_Box
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Purpose

This guide explains the current Tool_Box interface and the functionality implemented in version 0.29. It is written as a user and maintenance reference for lab, training, and other authorized security-testing environments.

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1. Program Overview

Tool_Box is a Bash-based, menu-driven wrapper for common Linux networking, enumeration, traffic-analysis, reporting, and lab support tools. Version 0.29 builds on the workspace and reporting system with faster global navigation, reorganized reference links, and temporary per-tool target/query overrides so users can move between tasks without repeatedly climbing through menu levels or overwriting the global target.

Feature	Behavior
Target context	Tracks a current IP address, subnet, selected ports, output directory, and active workspace.
Command builders	Builds commands for Nmap, Gobuster, FFUF, Feroxbuster, service enumeration, packet capture, and supporting utilities.
Safe execution	Displays the command before execution, asks for confirmation, marks elevated commands, and supports Dry-Run mode.
Global navigation	Nested-menu footer supports Back, Main Menu, Set Target/Data, Workspace, and Help shortcuts without discarding the current session state.
Results management	Can save output, search result trees, compare text output, keep command history, archive results, and generate Markdown reports.
Workspaces	Stores project state, notes, findings, and imported Nmap service inventories under a named workspace.
Software management	Detects supported tools and can install many of them on apt-based systems without mixing distribution repositories.
Desktop resources	Includes categorized Useful Links, custom bookmarks, clipboard support, and special Firefox launching from root sessions.

Scope note: Tool_Box should be used only on systems, labs, networks, and accounts you are authorized to test or administer.

2. Getting Started

2.1 Launching Tool_Box

```
chmod +x tool_box_v0.29.sh
./tool_box_v0.29.sh
```

Most actions should be performed as a normal desktop user. When an action needs elevated privileges, Tool_Box can use sudo for the command or start a nested root Tool_Box session from System & Configuration -> Privilege / User Session.

2.2 Default Startup Values

Setting	Default
Version	0.29
Target IP	127.0.0.1
Subnet	/32
Port selection	4444
Output directory	./results
Default threads	10
Default wordlist	SecLists common.txt when available; otherwise dirb/common.txt
Auto-Save Output	OFF
Verbose Mode	OFF
Dry-Run Mode	OFF
Color Output	ON

2.3 Suggested First-Time Setup

1. Open System & Configuration -> Tool_Box Diagnostics -> Run All Checks.
2. Open System & Configuration -> Install / Manage Software and scan for installed tools.
3. Open System & Configuration -> Settings and verify the default wordlist, threads, and output directory.
4. Create a Workspace from Target & Scanning -> Workspace / Project.
5. Set the target IP, subnet, and ports before running scans.

3. Interface and Global Behavior

Every main screen uses a persistent header so the current execution context remains visible while navigating menus.

```
=====
Tool_Box - <Menu Name> v0.29
=====
Target : <ip>/<subnet> Ports: <ports>
Output : <output directory>
Project: <workspace or None>
Modes  : Save ON/OFF Verbose ON/OFF Dry-Run ON/OFF
User   : <username> [standard] or root [ROOT]
```

Feature	Behavior
Target	The IP and subnet used by scans and enumeration actions.
Ports	The currently selected port list/range.
Output	The root directory used for Tool_Box result storage.
Project	The active workspace name when a workspace is loaded.
Save	Whether generic logged commands also write timestamped output files.
Verbose	Stored as a global mode and shown in the header; available to workflows that choose to use it.
Dry-Run	Previews commands and records them as Dry-Run without executing them.
User	Clearly marks standard-user versus root sessions.
Footer	Provides [O] Back, [M] Main Menu, [T] Set Target/Data, [W] Workspace, and [H] Help on nested menus.

3.1 Command Execution Safeguards

- **Command preview** - Tool_Box prints the full command before running it.
- **Confirmation** - Commands require explicit confirmation before execution.
- **Privilege warning** - Commands marked as elevated receive an additional warning/confirmation.
- **Dry-Run** - The command is previewed and recorded but is not executed.
- **Command history** - Run and result entries are persisted under the configured output root when possible.
- **Sensitive argument redaction** - SNMP community strings supplied with snmpwalk are redacted from command history.

3.2 Global Navigation Footer

Version 0.29 adds a navigation footer to menu prompts. The shortcuts are available from nested menus so the user can jump directly to common destinations without repeatedly selecting Back. The current target, ports, settings, output directory, privilege state, and active workspace remain in memory when using these shortcuts.

[O] Back [M] Main Menu [T] Set Target/Data [W] Workspace [H] Help

On the Main Menu, the footer omits Back and Main Menu because those actions are not useful at the top level. The available shortcuts are [T] Set Target/Data, [W] Workspace, and [H] Help.

- O - return one local menu level.
- M - return directly to the Main Menu from a nested menu.
- T - open Set Target / Data, then return to the Main Menu.
- W - open Workspace / Project, then return to the Main Menu.
- H - open the Quick Help / Workflow Guide, then return to the Main Menu.

4. Main Menu Architecture

Optimized Main Menu

Version 0.29 keeps the optimized six-category main menu while adding global footer shortcuts. The six categories remain intentionally broad; new functions should generally be placed inside an existing category instead of continually expanding the top-level menu.

Option	Function / Setting
1	Target & Scanning
2	Reconnaissance & Enumeration

Option	Function / Setting
3	Access & Exploitation
4	Analysis & Reporting
5	Utilities & Resources
6	System & Configuration
0	Exit

5. Target & Scanning

This section defines the current target and provides the project/workspace features that tie scans, notes, findings, services, and reports together.

Target & Scanning Menu

Option	Function / Setting
1	Workspace / Project
2	Set Target / Data
3	Scan Profiles
4	Import Nmap Results
5	Service-Aware Recommendations
0	Back

5.1 Workspaces / Projects

A workspace is a named project directory stored under the current output directory. It preserves the target state and provides separate files for notes, findings, and imported services.

Feature	Behavior
Create / Load	Creates a project directory or loads a previously saved workspace.
Save state	Stores workspace name, target IP/subnet, ports, wordlist, threads, output folder, and updated timestamp in workspace.conf.
Dashboard	Shows result file count/size, imported services, note count, and finding count.
Notes	Timestamped Markdown notes with user-defined categories such as General, To-Do, Credentials, or Host.
Findings / Loot	Timestamped TSV records. Suggested categories are host, service, username, credential, URL, file, hash, and other.
Nmap import	Parses Nmap XML, grepable output, or normal text into services.tsv; may update the current target and selected ports.
Service recommendations	Maps imported or selected ports/services to the most relevant Tool_Box area without automatically running scans.
Archive	Creates a timestamped .tar.gz copy of the current workspace.
Delete protection	Deletion is restricted to the workspace root and requires typing

Feature	Behavior
	the workspace name exactly.

Workspace / Project Menu

Option	Function / Setting
1	Create Workspace
2	Load Workspace
3	Save Current Workspace State
4	Workspace Dashboard
5	Notes
6	Findings / Loot Tracker
7	Import Nmap Results
8	Service-Aware Recommendations
9	List Workspaces
10	Archive Current Workspace
11	Unload Workspace
12	Delete Current Workspace
0	Back

5.2 Nmap Result Import

Import requires an active workspace and Python 3. The parser recognizes Nmap XML, .gnmap/grepable output, and normal text output. Open service entries are written to services.tsv with host, protocol, port, service, product, and version fields. If an IPv4 host and open ports are detected, Tool_Box updates the current target and selected ports to match the imported scan.

Common Port	Recommendation
21	FTP -> Service Enumeration
22	SSH -> Service Enumeration
25/465/587	SMTP -> Service Enumeration
53	DNS -> Network / DNS
80/443/8000/8008/8080/8081/8443/8888	HTTP(S) -> Web Enumeration
111	RPC -> Service Enumeration
139/445	SMB -> SMB / Windows
161/162	SNMP -> Service Enumeration
389/636	LDAP -> Service Enumeration
2049	NFS -> Service Enumeration
3306	MySQL -> Service Enumeration
3389	RDP -> SMB / Windows / Service Enumeration

Common Port	Recommendation
5432	PostgreSQL -> Service Enumeration

5.3 Set Target / Data

Set Data Menu

Option	Function / Setting
1	Set IP - validates an IPv4 address
2	Set Subnet - accepts /1 through /32
3	Set Port(s) - accepts individual ports, ranges, and comma/space-separated selections
4	Set Output Folder - creates/verifies a writable output root
0	Back

5.4 Scan Profiles

Scan Profiles Menu

Option	Function / Setting
1	Quick Recon - Nmap -T4 -F -sV
2	Full TCP Recon - Nmap -p- -sV -T4
3	Web Server - Nmap, WhatWeb, Curl headers, optional Gobuster
4	Windows / SMB - Nmap SMB scripts and SMBClient share listing
5	Linux Server - Nmap -sV -sC on selected ports
6	Custom - opens the Nmap builder
7	Scan Profiles Software
0	Back

5.5 Nmap Custom Builder

Nmap Builder

Option	Function / Setting
1	SYN Scan [-sS]
2	Connect Scan [-sT]
3	UDP Scan [-sU]
4	Version Detection [-sV] toggle
5	OS Detection [-O] toggle
6	Default Scripts [-sC] toggle
7	Skip Ping [-Pn] toggle

Option	Function / Setting
8	Fast Timing [-T4] toggle
9	Aggressive Scan [-A] toggle
10	Custom Options
11	Save Output toggle; uses timestamped Nmap output path
12	Run Command
13	View Nmap Man Page / Help
0	Back

The builder always uses the current Tool_Box target/subnet and selected ports unless custom options alter Nmap behavior.

6. Reconnaissance & Enumeration

Reconnaissance & Enumeration Hub

Option	Function / Setting
1	Reconnaissance
2	Web Enumeration
3	Network / DNS
4	SMB / Windows
5	Service Enumeration
6	Vulnerability Assessment
0	Back

6.1 Reconnaissance

Reconnaissance Menu

Option	Function / Setting
1	Nmap - Custom Builder
2	Host Discovery
3	Port Discovery
4	Service Detection
5	DNS Enumeration
6	WHOIS
7	Traceroute
8	Reconnaissance Software
0	Back

WHOIS and Traceroute support temporary custom IP/domain values. The custom value is local to that tool page and does not replace the global Tool_Box target. Both pages provide a reset option that returns the query to the current global target.

6.2 Web Enumeration

Web Enumeration Menu

Option	Function / Setting
1	Gobuster
2	FFUF
3	Feroxbuster
4	WhatWeb
5	Nikto
6	HTTP Headers (Curl)
7	TLS Certificate (OpenSSL)
8	Test HTTP/HTTPS
9	Web Enumeration Software
0	Back
Feature	Behavior
Gobuster	Protocol/HTTPS, redirects, expanded URLs, TLS validation, quiet mode, threads, extensions, wordlist, custom options, port, optional output file.
FFUF	Protocol, port, wordlist, threads, extensions, custom options, run/help.
Feroxbuster	Protocol, port, wordlist, threads, extensions, custom options, run/help.
WhatWeb	Protocol and port selection, then technology identification.
Nikto	Protocol and port selection, then web-server assessment.
Curl headers	Retrieves response headers for the selected protocol/port.
OpenSSL TLS viewer	Inspects a TLS certificate using a selected port and server name.
HTTP/HTTPS test	Simple connectivity/request test using Curl.

6.3 Network / DNS

Network / DNS Menu

Option	Function / Setting
1	DNS Lookup
2	Reverse DNS
3	WHOIS
4	Traceroute
5	ARP Scan
6	Test TCP Port (Netcat)

Option	Function / Setting
7	Local Interfaces
8	Routing Table
9	Network / DNS Software
0	Back

Custom Network/DNS Queries

Several Network / DNS tools can be pointed at a temporary query or host without changing the global target. DNS Lookup accepts a hostname/DNS name, Reverse DNS accepts a custom IP address, WHOIS accepts an IP or domain, Traceroute accepts an IP or domain, and Netcat TCP Test accepts both a custom IP/domain and port. Leaving or resetting the tool returns normal operations to the global target context.

6.4 SMB / Windows

SMB / Windows Menu

Option	Function / Setting
1	SMB Enumeration (enum4linux-ng)
2	List SMB Shares
3	SMB Client
4	Nmap SMB Information Scripts
5	NetBIOS Information
6	SMB / Windows Software
0	Back

6.5 Service Enumeration

Service Enumeration Menu

Option	Function / Setting
1	SSH
2	FTP
3	SMTP
4	SNMP
5	LDAP
6	NFS
7	RPC
8	Database Services
9	Service Enumeration Software
0	Back
Feature	Behavior

Option	Function / Setting
SSH	Port selection, Nmap version detection, ssh-keyscan, Nmap SSH algorithms/host-key checks.
FTP	Port selection, banner/version detection, anonymous FTP/system info checks.
SMTP	Port selection, banner, SMTP command enumeration, version detection.
SNMP	Port, supplied community string, OID, and snmpwalk. Community strings are redacted from history.
LDAP	Port, Base DN, RootDSE query, selected-base query.
NFS	showmount exports and rpcinfo services.
RPC	Registered RPC program listing.
Database Services	Port selection and Nmap service detection for common database services.

6.6 Vulnerability Assessment

Vulnerability Assessment Menu

Option	Function / Setting
1	Set Assessment Port
2	Safe NSE Script Scan
3	TLS Cipher Assessment
4	HTTP Security Headers
5	SMB Security Configuration
6	Service Detection + Default Scripts
7	Vulnerability Assessment Software
8	View Nmap Man Page / Help
0	Back

This section is designed around enumeration and assessment checks rather than automatic exploitation.

7. Access & Exploitation

Access & Exploitation Menu

Option	Function / Setting
1	Pass / Hash Cracking
2	Exploit
0	Back

7.1 Pass / Hash Cracking

Pass / Hash Cracking Menu

Option	Function / Setting
1	Hash Identification Helper
2	Text / Encoding Tools
0	Back

In v0.29 this category currently provides analysis helpers rather than automated password cracking workflows.

7.2 Exploit / Metasploit

Metasploit Menu

Option	Function / Setting
1	Launch msfconsole
2	Show Version
3	View Metasploit Man Page / Help
0	Back

Tool_Box launches the local Metasploit console but does not install Metasploit automatically.

8. Analysis & Reporting

Analysis & Reporting Menu

Option	Function / Setting
1	Traffic Analysis
2	Results / Reporting
0	Back

8.1 Traffic Analysis

Traffic Analysis Menu

Option	Function / Setting
1	Start tcpdump Capture
2	Read Capture
3	List Capture Interfaces
4	Capture by Host
5	Capture by Port
6	TShark Analysis
7	Traffic Analysis Software
0	Back

Option	Function / Setting
Feature	Behavior
tcpdump capture	Select interface, packet count, optional port, then start a capture.
Read capture	Read a selected PCAP with tcpdump.
Capture by host/port	Builds focused packet-capture filters.
TShark	Selects a PCAP and optional display filter for command-line analysis.

8.2 Results / Reporting

Results / Reporting Menu

Option	Function / Setting
1	Results Dashboard
2	List Targets with Results
3	List Recent Result Files
4	View a Text Result
5	Show Current Target Result Directory
6	Search Current Target Results
7	Compare Two Text Results
8	Generate Markdown Target Report
9	Archive Current Target Results
10	Delete a Result File
11	View Tool_Box Command History
12	Search Tool_Box Command History
0	Back
Feature	Behavior
Results Dashboard	Shows result file count, category count, disk size, command-history count, and active workspace statistics.
Search	Recursively searches text results and also includes workspace notes/findings when a workspace is loaded.
Compare	Uses unified diff output for two selected text files.
Markdown report	Creates a report containing target metadata, imported services, findings, notes, and the first 40 lines of each text result. Binary/archive content is not inlined.
Archive	Creates a timestamped .tar.gz archive of the current target result directory.
Delete protection	Refuses to delete files outside the configured output root and requires confirmation.
History	Records commands, Dry-Run previews, exit status, target, and

Option	Function / Setting
	output file path where applicable.

9. Utilities & Resources

Utilities & Resources Menu

Option	Function / Setting
1	Command-Line Utilities
2	Network Information / VPN Status
3	Text / Encoding Tools
4	Port Reference
5	Useful Links
6	Quick Help / Workflow Guide
0	Back

9.1 Command-Line Utilities

Command-Line Utilities Menu

Option	Function / Setting
1	Curl
2	Wget
3	Netcat TCP Test
4	JQ
5	OpenSSL
6	Utilities Software
0	Back

9.2 Network Information / VPN Status

Network Information Menu

Option	Function / Setting
1	Local Interfaces / Routes / DNS
2	VPN / Tunnel Status
3	Public IP (external request)
0	Back

The network page displays local interface information, routing data, neighbor information when available, and resolver information. The VPN page searches for common tunnel-style interfaces such as tun, tap, and wg. Public IP performs an external request and therefore requires network access.

9.3 Text / Encoding Tools

Text / Encoding Menu

Option	Function / Setting
1	Base64 Encode
2	Base64 Decode
3	Hex Encode
4	Hex Decode
5	URL Encode
6	URL Decode
7	SHA-256 Text Hash
8	Hash Identification Helper
0	Back

9.4 Port Reference

The Port Reference accepts a port number and displays the common service description known to Tool_Box. It is intended as a quick navigation/reference aid when deciding which enumeration menu to use.

9.5 Useful Links

Useful Links Categories

Option	Function / Setting
1	Shell / Session Tools
2	Linux / Windows Privilege References
3	Data / Encoding / Analysis
4	Web Security
5	Recon / Vulnerability References
6	Payloads / Wordlists / Cheat Sheets
7	Training
8	Custom Bookmarks
0	Back
Feature	Behavior
Shell / Session Tools	Penelope and Reverse Shell Generator.
Privilege References	GTFOBins, LOLBAS, and HackTricks. The older dedicated GTFOBins-less shortcut is not included.
Data / Encoding / Analysis	CyberChef and VirusTotal.
Web Security	PortSwigger Web Security Academy, OWASP Cheat Sheet Series, and OWASP Web Security Testing Guide.
Recon / Vulnerabilities	crt.sh, Nmap Reference Guide, Nmap NSE Documentation,

Option	Function / Setting
	Exploit Database, MITRE ATT&CK, CVE.org, and NIST NVD.
Payloads / Wordlists / Cheat Sheets	PayloadsAllTheThings, SecLists, and PacketLife Cheat Sheets.
Training	TryHackMe and PortSwigger Web Security Academy.
Custom Bookmarks	Stores user-added HTTP/HTTPS bookmarks in ~/.tool_box_links.tsv with mode 600 when possible.

CyberChef is located under Data / Encoding / Analysis together with VirusTotal. GTFOBins is provided as the general privilege-reference site; the older dedicated "GTFOBins - less" shortcut is no longer included. Certificate Transparency (crt.sh) is grouped under Recon / Vulnerability References.

9.6 Link Opening Behavior

Feature	Behavior
Default Browser	Uses xdg-open when available.
Firefox	Detects firefox or firefox-esr.
Chrome / Chromium	Detects common Chrome and Chromium command names; the browser does not have to be installed to use other options.
Copy URL	Uses wl-copy, xclip, or xsel when one is available.
Root session	Adds "Open Firefox as User". Tool_Box prompts for the desktop username, validates the user, discovers the user GUI session, and launches Firefox under that account.
GUI environment discovery	Attempts to recover XDG_RUNTIME_DIR, D-Bus, Wayland/X11 display, and Xauthority information from the selected user session.
Browser diagnostics	Displays startup errors when a browser fails instead of silently discarding them.

10. System & Configuration

System & Configuration Menu

Option	Function / Setting
1	Install / Manage Software
2	Settings
3	Tool_Box Diagnostics
4	Privilege / User Session
5	About / Version / Update Check
0	Back

10.1 Install / Manage Software

Software Dependencies Menu

Option	Function / Setting
1	Scan for Installed
2	Install / Manage Supported Software
3	Install Missing
4	Repository Troubleshooter
5	Update Package Lists
6	Show Supported Software Versions
7	Show Missing Supported Software
0	Back

The installer is designed for apt-based systems. Tool_Box detects the current distribution and only enables repositories that belong to that operating system. It specifically avoids mixing Kali repositories into Ubuntu/Debian systems or the reverse.

Repository Troubleshooter

Option	Function / Setting
1	Show OS / Active Repositories
2	Enable Standard Repositories for This OS
3	Update Package Lists
4	Check Tool Package Availability
5	Run APT Diagnostics
0	Back

10.2 Diagnostics

Tool_Box Diagnostics Menu

Option	Function / Setting
1	Run All Checks
2	Software Status
3	Output Folder Check
4	Default Wordlist Check
5	Network Interfaces / Routes
6	Configuration Check
7	Privilege / Package Manager Info
8	Bash Syntax Check
9	APT / Repository Troubleshooter
10	Desktop / Browser / Clipboard Check

Option	Function / Setting
0	Back

10.3 Privilege / User Session

A standard-user session can start a nested root instance using sudo. The current target, ports, output path, wordlist, threads, modes, and workspace context are passed into the root child. When that child exits, the original user session resumes with its in-memory state intact.

Feature	Behavior
Standard user	Menu offers "Start Tool_Box as Root" when sudo is available.
Nested root	Header changes to root [ROOT], and the Privilege menu offers a return to the parent Tool_Box session.
Direct root launch	Tool_Box recognizes that no parent user session exists.
Elevated commands	Individual privileged commands are marked and require confirmation before execution.

10.4 About / Version / Update Check

About Menu

Option	Function / Setting
1	Open Repository
2	Copy Repository URL
3	Check Repository Version
4	Privilege Guide
0	Back

The update checker compares the version string in the local script with the repository copy. It does not overwrite Tool_Box automatically.

11. Settings and Persistent Configuration

Settings Menu

Option	Function / Setting
1	Default Wordlist
2	Default Threads
3	Output Directory
4	Auto-Save Output toggle
5	Verbose Mode toggle
6	Dry-Run Mode toggle
7	Color Output toggle
8	Save Configuration

Option	Function / Setting
9	Load Configuration
10	Reset Defaults
11	Settings Profiles
0	Back
Feature	Behavior
Default Wordlist	Used by Gobuster, FFUF, Feroxbuster, and profiles that need a content wordlist.
Default Threads	Used as the starting thread count for web-enumeration tools and relevant profiles.
Output Directory	Root for target results, command history, workspaces, and archives.
Auto-Save Output	When ON, run_command_logged saves timestamped output files in target/category directories.
Verbose Mode	Global preference shown in the header and saved in configuration/profiles.
Dry-Run Mode	Previews commands and records them without execution.
Color Output	Controls ANSI styling; color is also suppressed automatically for noninteractive terminals, TERM=dumb, or NO_COLOR.
Save/Load Configuration	Stores or reloads the main settings from ~/.tool_box.conf.
Reset Defaults	Restores session defaults without deleting saved profiles/config files.
Settings Profiles	Saves multiple named configuration presets in ~/.tool_box_profiles/.

Settings Profiles Menu

Option	Function / Setting
1	Save Current Settings as Profile
2	Load Profile
3	Delete Profile
4	List Profiles
0	Back

11.1 Values Saved in Configuration and Profiles

```

default_threads=<number>
default_wordlist=<path>
output_folder=<path>
auto_save_output=true|false
verbose_mode=true|false
dry_run_mode=true|false
color_enabled=true|false

```

12. Output, Workspaces, and File Locations

Item	Location / Pattern
Main configuration	~/.tool_box.conf
Settings profiles	~/.tool_box_profiles/<profile>.conf
Custom Useful Links	~/.tool_box_links.tsv
Command history	<output_folder>/tool_box_command_history.log
Target results	<output_folder>/<target>/<category>/<target>-<timestamp>-<tool>.<ext>
Workspaces	<output_folder>/workspaces/<workspace>/
Workspace state	workspace.conf
Workspace notes	notes.md
Workspace findings	findings.tsv
Imported services	services.tsv
Workspace archives	<output_folder>/workspace_archives/<workspace>-<timestamp>.tar.gz
Target archives	<output_folder>/archives/<target>-<timestamp>.tar.gz
Markdown reports	Inside the current target result directory as report-<timestamp>.md

12.1 Example Result Layout

```

results/
  tool_box_command_history.log
  10.10.10.10/
    nmap/
    web/
    profiles/
    traffic/
    ...
    report-YYYY-MM-DD_HH-MM-SS.md
  workspaces/
    Example_Project/
      workspace.conf
      notes.md
      findings.tsv
      services.tsv
  archives/
  workspace_archives/

```

13. Privileges and Browser Launching

Privilege Level	Typical Use
[USER]	Most web, DNS, SMB client, service-enumeration, text-processing, and reporting actions.
[ROOT OPTIONAL]	Some Nmap scan modes can benefit from raw-socket privileges/capabilities.
[ROOT REQUIRED]	Packet capture, some ARP/network discovery operations, and system package/repository changes.

The root-session workflow is intentionally nested rather than replacing the user process. This allows the user to perform an elevated task and return to the original Tool_Box session. For graphical browser launches from a root Tool_Box, use the explicit Firefox-as-user option so Firefox runs inside the logged-in desktop account rather than as root.

14. Supported Software

Tool	APT Package	Automatic Install Support
Nmap	nmap	Yes
Gobuster	gobuster	Yes
Curl	curl	Yes
Wget	wget	Yes
Netcat	netcat-openbsd	Yes
Dig	dnsutils	Yes
Whois	whois	Yes
Traceroute	traceroute	Yes
JQ	jq	Yes
OpenSSL	openssl	Yes
tcpdump	tcpdump	Yes
TShark	tshark	Yes
FFUF	ffuf	Yes
Feroxbuster	feroxbuster	Yes
WhatWeb	whatweb	Yes
Nikto	nikto	Yes
SMBClient	smbclient	Yes
enum4linux-ng	enum4linux-ng	Yes
arp-scan	arp-scan	Yes
ip	iproute2	Yes
OpenSSH Client	openssh-client	Yes
SNMPWalk	snmp	Yes
LDAPSearch	ldap-utils	Yes
Showmount	nfs-common	Yes
RPCInfo	rpcbind	Yes
Metasploit	N/A	No - launcher/documentation only

14.1 Section-Specific Software Pages

Reconnaissance, Web Enumeration, Network/DNS, SMB/Windows, Traffic Analysis, Utilities, Scan Profiles, Service Enumeration, and Vulnerability Assessment each expose a focused software status page. Selecting a tool opens a details page with install/repair, version, man/help, and repository troubleshooting options when applicable.

15. Recommended Workflow

1. Create or load a Workspace under Target & Scanning.
2. Set the target IP, subnet, ports, output directory, default wordlist, and threads.
3. Run an Nmap builder scan or a predefined Scan Profile.
4. If you saved Nmap output, import the XML, grepable, or normal text result into the workspace.
5. Review Service-Aware Recommendations to decide which enumeration category to open next.
6. Use Web, SMB, Service, DNS, or Vulnerability Assessment menus to investigate discovered services.
7. Add important observations to Workspace Notes and structured items to Findings / Loot Tracker.
8. Use Results / Reporting to search output, inspect command history, compare scans, and generate a Markdown report.
9. Archive the target results or workspace when you want a portable snapshot.

16. Troubleshooting Quick Reference

Problem	What to Check
Too many menu levels / hard to navigate	Use the global footer shortcuts: M returns to Main Menu, T opens Set Target/Data, W opens Workspace, H opens Quick Help, and 0 goes back one level.
Firefox will not open while Tool_Box is root	Use Useful Links -> select a link -> Open Firefox as User. Enter the non-root username that owns the active desktop session. Run Desktop / Browser / Clipboard Diagnostics if it still fails.
A tool shows Missing	Open System & Configuration -> Install / Manage Software. Scan installed tools, then install/manage the missing package or use the repository troubleshooter.
Output is not being saved	Check whether Auto-Save Output is ON. Also run Diagnostics -> Output Folder Check and verify the target tool has an explicit save toggle where applicable.
Nmap import fails	Load/create a workspace, verify the scan file exists, and confirm Python 3 is installed. Supported formats are XML, grepable, and normal Nmap text.
Wrong target after loading a workspace	Workspace state intentionally restores the saved target/subnet/ports. Change the target and save the workspace state if the project has moved to a new host.
Root-only action fails	Use Privilege / User Session to start a nested root Tool_Box or allow the specific elevated command to run with sudo when prompted.
Colors look wrong	Disable Color Output, set NO_COLOR, or use a normal interactive terminal.
APT package unavailable	Use Repository Troubleshooter -> Show OS / Active Repositories and Check Tool Package Availability. Tool_Box will not mix repositories from different distributions.
Clipboard unavailable	Install/use a supported helper such as wl-copy, xclip, or xsel; the URL remains visible even if copying fails.
Need to verify the script itself	Run Diagnostics -> Bash Syntax Check.

Appendix A. Complete Menu Reference

This appendix provides a compact menu-by-menu reference. Some action pages, such as individual software details or service-specific builders, are summarized here rather than repeating every help command described in the main guide. Unless otherwise noted, nested menus also display the v0.29 navigation footer: [0] Back, [M] Main Menu, [T] Set Target/Data, [W] Workspace, [H] Help.

A.1 Main and Category Hubs

Main Menu

Option	Function / Setting
1	Target & Scanning
2	Reconnaissance & Enumeration
3	Access & Exploitation
4	Analysis & Reporting
5	Utilities & Resources
6	System & Configuration
0	Exit

Target & Scanning

Option	Function / Setting
1	Workspace / Project
2	Set Target / Data
3	Scan Profiles
4	Import Nmap Results
5	Service-Aware Recommendations
0	Back

Reconnaissance & Enumeration

Option	Function / Setting
1	Reconnaissance
2	Web Enumeration
3	Network / DNS
4	SMB / Windows
5	Service Enumeration
6	Vulnerability Assessment
0	Back

Access & Exploitation

Option	Function / Setting
1	Pass / Hash Cracking

Option	Function / Setting
2	Exploit
0	Back

Analysis & Reporting

Option	Function / Setting
1	Traffic Analysis
2	Results / Reporting
0	Back

Utilities & Resources

Option	Function / Setting
1	Command-Line Utilities
2	Network Information / VPN Status
3	Text / Encoding Tools
4	Port Reference
5	Useful Links
6	Quick Help / Workflow Guide
0	Back

System & Configuration

Option	Function / Setting
1	Install / Manage Software
2	Settings
3	Tool_Box Diagnostics
4	Privilege / User Session
5	About / Version / Update Check
0	Back

A.2 Target, Workspace, and Profiles

Set Data

Option	Function / Setting
1	Set IP
2	Set Subnet
3	Set Port(s)
4	Set Output Folder
0	Back

Workspace / Project

Option	Function / Setting
1	Create Workspace
2	Load Workspace
3	Save Current Workspace State
4	Workspace Dashboard
5	Notes
6	Findings / Loot Tracker
7	Import Nmap Results
8	Service-Aware Recommendations
9	List Workspaces
10	Archive Current Workspace
11	Unload Workspace
12	Delete Current Workspace
0	Back

Workspace Notes

Option	Function / Setting
1	Add Note
2	View Notes
3	Edit Notes in Editor
0	Back

Findings / Loot Tracker

Option	Function / Setting
1	Add Finding
2	View Findings
3	Search Findings
0	Back

Scan Profiles

Option	Function / Setting
1	Quick Recon
2	Full TCP Recon
3	Web Server
4	Windows / SMB
5	Linux Server

Option	Function / Setting
6	Custom (Open Nmap Builder)
7	Scan Profiles Software
0	Back

A.3 Reconnaissance and Web Enumeration

Reconnaissance

Option	Function / Setting
1	Nmap - Custom Builder
2	Host Discovery
3	Port Discovery
4	Service Detection
5	DNS Enumeration
6	WHOIS
7	Traceroute
8	Reconnaissance Software
0	Back

Web Enumeration

Option	Function / Setting
1	Gobuster
2	FFUF
3	Feroxbuster
4	WhatWeb
5	Nikto
6	HTTP Headers (Curl)
7	TLS Certificate (OpenSSL)
8	Test HTTP/HTTPS
9	Web Enumeration Software
0	Back

Network / DNS

Option	Function / Setting
1	DNS Lookup
2	Reverse DNS
3	WHOIS

Option	Function / Setting
4	Traceroute
5	ARP Scan
6	Test TCP Port (Netcat)
7	Local Interfaces
8	Routing Table
9	Network / DNS Software
0	Back

A.4 SMB, Services, and Assessment

SMB / Windows

Option	Function / Setting
1	SMB Enumeration (enum4linux-ng)
2	List SMB Shares
3	SMB Client
4	Nmap SMB Information Scripts
5	NetBIOS Information
6	SMB / Windows Software
0	Back

Service Enumeration

Option	Function / Setting
1	SSH
2	FTP
3	SMTP
4	SNMP
5	LDAP
6	NFS
7	RPC
8	Database Services
9	Service Enumeration Software
0	Back

Vulnerability Assessment

Option	Function / Setting
1	Set Assessment Port

Option	Function / Setting
2	Safe NSE Script Scan
3	TLS Cipher Assessment
4	HTTP Security Headers
5	SMB Security Configuration
6	Service Detection + Default Scripts
7	Vulnerability Assessment Software
8	View Nmap Man Page / Help
0	Back

A.5 Analysis, Reporting, and Utilities

Traffic Analysis

Option	Function / Setting
1	Start tcpdump Capture
2	Read Capture
3	List Capture Interfaces
4	Capture by Host
5	Capture by Port
6	TShark Analysis
7	Traffic Analysis Software
0	Back

Results / Reporting

Option	Function / Setting
1	Results Dashboard
2	List Targets with Results
3	List Recent Result Files
4	View a Text Result
5	Show Current Target Result Directory
6	Search Current Target Results
7	Compare Two Text Results
8	Generate Markdown Target Report
9	Archive Current Target Results
10	Delete a Result File
11	View Tool_Box Command History
12	Search Tool_Box Command History

Option	Function / Setting
0	Back

Command-Line Utilities

Option	Function / Setting
1	Curl
2	Wget
3	Netcat TCP Test
4	JQ
5	OpenSSL
6	Utilities Software
0	Back

Network Information

Option	Function / Setting
1	Local Interfaces / Routes / DNS
2	VPN / Tunnel Status
3	Public IP (external request)
0	Back

Text / Encoding Tools

Option	Function / Setting
1	Base64 Encode
2	Base64 Decode
3	Hex Encode
4	Hex Decode
5	URL Encode
6	URL Decode
7	SHA-256 Text Hash
8	Hash Identification Helper
0	Back

A.6 Useful Links and Administration

Useful Links

Option	Function / Setting
1	Shell / Session Tools
2	Linux / Windows Privilege References

Option	Function / Setting
3	Data / Encoding / Analysis
4	Web Security
5	Recon / Vulnerability References
6	Payloads / Wordlists / Cheat Sheets
7	Training
8	Custom Bookmarks
0	Back

Software Dependencies

Option	Function / Setting
1	Scan for Installed
2	Install / Manage Supported Software
3	Install Missing
4	Repository Troubleshooter
5	Update Package Lists
6	Show Supported Software Versions
7	Show Missing Supported Software
0	Back

Settings

Option	Function / Setting
1	Default Wordlist
2	Default Threads
3	Output Directory
4	Auto-Save Output
5	Verbose Mode
6	Dry-Run Mode
7	Color Output
8	Save Configuration
9	Load Configuration
10	Reset Defaults
11	Settings Profiles
0	Back

Diagnostics

Option	Function / Setting
1	Run All Checks
2	Software Status
3	Output Folder Check
4	Default Wordlist Check
5	Network Interfaces / Routes
6	Configuration Check
7	Privilege / Package Manager Info
8	Bash Syntax Check
9	APT / Repository Troubleshooter
10	Desktop / Browser / Clipboard Check
0	Back

About

Option	Function / Setting
1	Open Repository
2	Copy Repository URL
3	Check Repository Version
4	Privilege Guide
0	Back

Appendix B. v0.29 Feature Summary

- Named workspaces/projects with load/save/archive/delete support.
 - Timestamped workspace notes and structured findings/loot tracking.
 - Nmap XML, grepable, and normal-text result import into a service inventory.
 - Service-aware recommendations that point to relevant Tool_Box sections.
 - Workspace and results dashboards.
 - Result-tree search and persistent command-history search.
 - Markdown reports that incorporate imported services, findings, notes, and result previews.
 - Settings profiles for reusable configuration presets.
 - Network interface, VPN/tunnel, public-IP, encoding, hash-identification, and port-reference utilities.
 - Categorized Useful Links with separate Data / Encoding / Analysis references plus user-defined custom bookmarks.
 - Default-browser and clipboard integration.
 - Root-session Firefox-as-user launcher with desktop-session environment discovery.
 - Nested root Tool_Box session that carries the active target, settings, and workspace context.
 - Desktop/browser/clipboard diagnostics and privilege guidance.
 - Repository version check that does not automatically overwrite the local script.
-
- Global navigation footer shortcuts: Back, Main Menu, Set Target/Data, Workspace, and Help.
 - Temporary per-tool targets/queries for WHOIS, Traceroute, DNS Lookup, Reverse DNS, and Netcat TCP Test.
 - Useful Links reorganized with a dedicated Data / Encoding / Analysis category; CyberChef moved there and the dedicated GTFOBins-less shortcut removed.